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A Input Templates

The selected relations include: `player` — 'birth place', 'birth date', 'position', 'nationality'; `movie` — 'director', 'release date', 'genre', 'country'; `city` — 'country', 'first mayor', 'founded date', 'climate type'; and `song` — 'artist', 'album', 'release date', 'language'.

Initially, we constructed the input templates using the relations described in Ferrando et al. (2024) for four categories: `football player`, `movie`, `city`, and `song`, we call this set of relations as `relations1`. The specific relationships extracted for each category are as follows:

- `player`: birthplace, birth date, teams played.
- `movie`: director, screenwriter, release date, genre, duration, cast.
- `city`: country, population, elevation, coordinates.
- `song`: artist, album involvement, publication year, genre.

Since the number of relations is not balanced for the categories and some relation answers have non-trivial modality, e.g. coordinates of a city, we propose a new unified dataset, that is balanced in the sense of number of features and standard output modalities, we call this set of relations `relations2`. The following relations shape the `unified` dataset:

- `player`: birthplace, birthdate, position, nationality.
- `movie`: director, release date, genre, production country.
- `city`: country, population, founded date, timezone.
- `song`: artist, album label, release date, language.

Afterwards, we hand-craft templates from the quadruples to form statements. For example, "The movie Inception was directed by *director* Christopher Nolan". Since the expected answer for some relations can be ambiguous as in "The player Michael Jordan was born in *city of*..." where it is unclear whether the response should be a location or a year—we incorporate "hints" at the end of each relation.

We experimented with four input templates (see Table 3) using the `relations2` set, aiming to eliminate spurious correlations and isolate the self-awareness signal captured by linear probes. Notably, only `template2` consistently captures the self-awareness signal without interference from confounding factors.

Template Name	**Sample Sentence**
`template1`	"The player Youri Djorkaeff was born in the city of"
`template1_const_end`	"The player Youri Djorkaeff's birth city is"
`template2`	"The city of birth for the player Youri Djorkaeff is"
`template2_balanced`	"The city of birth for the player Youri Djorkaeff is"

Table 3: Input templates using the entity type "player" and entit name "Youri Djorkaeff"

`template1`: Places the entity type and name at the beginning of the sentence and ends with variable tokens. For example: *"The <entity_type> <entity_name> was born in the city of..."*

`template1_const_end`: Similar to `template1`, it places the entity type and name at the beginning of the sentence but always ends with the fixed token *"is"*. For example: *"The <entity_type> <entity_name>'s birth city is..."*

`template2`: In contrast to `template1`, it places the entity type and name at the end of the prompt. Like `template1_const_end`, it also ends with the token *"is"*. For example: *"The city of birth for the <entity_type> <entity_name> is..."*

`template2_balanced`: Further improvement of `template2` to have balanced number of known and forgotten samples across relations per category.

See the full set of templates from `template2_balanced` that were used in the experiments Table 3.

B Sample Distribution Across Entity Types

We present the full distribution of known and forgotten samples using `template2_balanced` for Gemma 2 2B with $k = 500$ and $l = 0.3$ parameters in Table 4 and for Pythia 12B in Table 5.

Table 4: Distribution of known and forgotten samples across entity categories for the Gemma 2 2B model, using top-$k = 500$ and bottom-$l = 0.3$ thresholds.

Category	**Known**	**Forgotten**	**Subset Total**
Player	1286	2922	4208
Movie	3602	1810	5412
City	1017	541	1558
Song	1475	1995	3470
Total	7380	7268	14648

Table 5: Distribution of known and forgotten samples across entity categories for the Pythia 12B model, using top-$k = 500$ and bottom-$l = 0.3$ thresholds.

Category	**Known**	**Forgotten**	**Subset Total**
Player	657	3551	4208
Movie	3602	1810	5412
City	574	984	1558
Song	538	2932	3470
Total	5371	9277	14648

C Separation Scores

The latent separation scores for Gemma 2 9B using SAE activations (Figure 9) and Linear Probe activations (Figure 10).

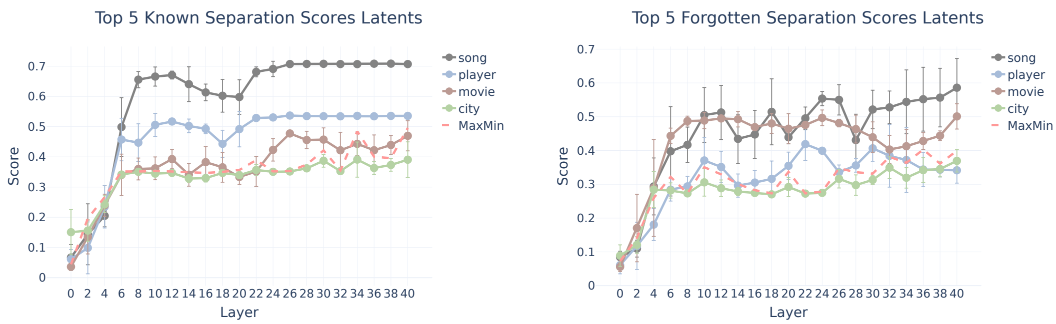

Figure 9: Latent separation scores using SAE activations on Gemma 2 9B. Left: **known** entities. Right: **forgotten** entities.

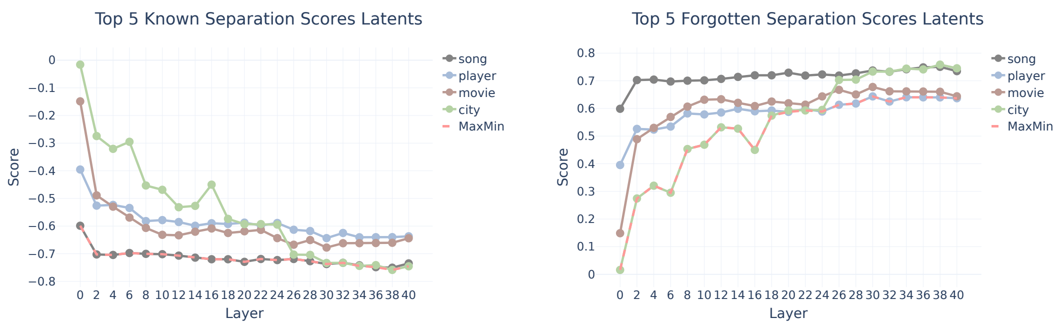

Figure 10: Latent separation scores using Linear Probe activations on Gemma 2 9B. Left: **known** entities. Right: **forgotten** entities.

D (k,l) Pair Impact on Probe Behavior

The figures in this section illustrate how varying the (k, l) parameters—representing the number of known and forgotten samples, respectively—affects linear probe performance and class balance outcomes across different model scales. Figure 11 presents the test and train accuracy gains over a random baseline for the Gemma 2 2B and Pythia 12B models. Notably, we observe that increases in k (the number of known samples) generally correspond to higher accuracy gains, particularly when l (the number of forgotten samples) remains low. This trend is more pronounced in larger models, consistent with their greater capacity to capture and retain class-discriminative features.

Figures 13 and 12 explore the implications of (k, l) settings on class balance, defined as the ratio of known to forgotten samples, for Pythia and Gemma 2 model families, respectively. The heatmaps indicate that this ratio grows with increasing k and decreasing l, with larger models showing a more marked divergence between known and forgotten categories. These findings highlight the sensitivity of probe performance and interpretability metrics to sampling configurations, underscoring the importance of systematic calibration of (k, l) pairs when designing probing protocols.